**R15**

**Code No: 123BR**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech II Year I Semester Examinations, November/December - 2018**
**BASIC ELECTRICAL ENGINEERING**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

**PART- A**

**(25 Marks)**

1.a) When three capacitors of 5μF each are connected in series, find its equivalent Capacitance? [2]
b) State Kirchhoff's laws with an example. [3]
c) Define form factor and peak factor. [2]
d) Write the significance of 'j' operator in ac circuits. [3]
e) What is meant by eddy current losses in a transformer? [2]
f) What are the properties of ideal transformer? [3]
g) A 3-phase induction motor is wound for 4 poles and is supplied with 415 V, 50 Hz system, Find its synchronous speed. [2]
h) What are the difference between long shunt and short shunt DC compound motor. [3]
i) Why controlling torque is necessary in an indicating instruments? [2]
j) Write the classification of instruments. [3]

**PART-B**

**(50 Marks)**

2.a) Discuss the classification of sources in detail.
b) State and explain Super position theorem. [5+5]

**OR**

3.a) Determine the current flowing through 15 Ω, 30 Ω and 40 Ω resistors if the applied DC voltage is 220 V. Also find power dissipated in 10 Ω resistor shown in figure 1.

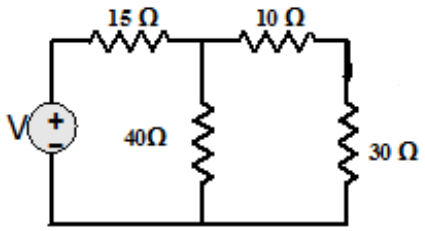



**Figure: 1**

b) Obtain the relation between star to delta transformation. [5+5]

4.a) What is a phasor diagram? Develop a phasor diagram for series RL circuit connected across a.c. supply.

b) Determine RMS value of the voltage defined by $v(t) = 5 + 5\sin\left(314t + \frac{\pi}{6}\right)$. [5+5]

**OR**

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5.a) Find the average value and form factor of the following voltage waveform shown in figure 2.

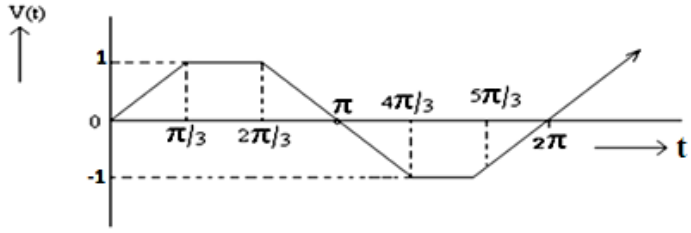



**Figure: 2**

b) Obtain the phase relations between voltage and current in a pure capacitive circuit excited with 1-ϕ ac supply. [6+4]

6.a) Discuss the main constructional components of a single phase Transformer.

b) Define efficiency and regulation in a Transformer. Give their formulae. [5+5]

**OR**

7.a) Discuss the practical transformer on no load condition.

b) A 1-ϕ transformer has 500 turns in the primary winding and cross sectional area of the core is 80 $cm^2$ and the primary winding is connected to a 50 Hz, 230 V supply. Calculate maximum flux induced in it. [5+5]

8.a) Derive the EMF equation of a DC Generator.

b) What is the importance of back EMF in a DC machine? [6+4]

**OR**

9.a) Describe the principle of operation of 3-ϕ induction motor.

b) Explain the following losses in a DC machine:
i) Copper losses ii) Iron losses iii) Mechanical Losses. [5+5]

10. Explain the working of PMMC instrument with a neat sketch. Also write its merits and demerits. [10]

**OR**

11.a) Describe the essential features of measuring instruments.

b) Write a short note on damping in measuring systems. [5+5]

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